

Program : Diploma in Chemical Engineering	
Course Code : 4071	Course Title: Process Heat Transfer
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To impart the knowledge on different modes of heat transfer, construction and working of different heat exchanging equipment in industries.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basic Differentiation and Integration		Mathematics I	1
		Mathematics II	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the mechanism of heat transfer through conduction and convection.	12	Applying
CO2	Summarise the heat transfer in fluids and radiation heat transfer.	17	Applying
CO3	Summarise heat exchangers.	16	Applying
CO4	Summarise evaporators.	15	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Interpret the mechanism of heat transfer through conduction and convection.		
M1.01	Outline the basic modes of heat transfer.	1	Understanding
M1.02	Interpret the mechanism of heat transfer through conduction.	6	Applying
M1.03	Interpret the mechanism of heat transfer through convection	3	Applying
M1.04	Interpret overall heat transfer coefficient.	2	Applying
Contents: Heat transfer – Modes of heat transfer, definitions. Heat transfer by conduction in solids – Steady state and unsteady state flow – definition - Units of heat flow - Fourier's law of conduction – Rate equation for heat flow – Steady state heat flow - Conduction through single wall – Derivation of equation and simple problems – Thermal conductivity – Units - Steady state conduction through composite wall in series - Derivation of equation and problems - Steady state conduction through cylindrical wall and spherical wall derivation – Problems - Economic thickness of insulation – Commonly used insulating materials. Theory of convection – Film concept of heat transfer temperature gradient in forced convection – Derivation of overall heat transfer coefficient from individual heat transfer coefficient – Simple problems. Compute overall heat transfer coefficient – Simple problems.			
CO2	Summarise the heat transfer in fluids and radiation heat transfer.		
M2.01	Interpret the mechanism of heat transfer by natural and forced convection.	6	Applying
M2.02	Outline the mechanism of heat transfer in boiling liquid.	4	Understanding

M2.03	Outline condensation.	2	Understanding
M2.04	Interpret radiation heat transfer.	5	Applying
	Series Test – I		
Contents: Mechanism of natural convection – Mechanism of Forced convection - Heat transfer by forced convection inside tubes – Heat transfer to fluid without phase change – Dimensionless equation-Reynolds number – Nusselt number -Prandtl number- Grashof number - Film coefficients in pipes laminar flow - sieder –Tate equation Film coefficient in turbulent flow – Dittus Boelters Equation – Simple problems. Heat transfer in boiling liquids – Flash boiling - Sub cooled boiling – Saturated boiling – Regimes of boiling. Condensation – Types of condensation Radiation – Elementary idea of black body – Gray body - Emissivity – Emissive power – Laws of radiation – Kirchoffs law, Planks law, Stefan Boltzmann law, Weins displacement law – Simple problems – Combined heat losses by convection and radiation			
CO3	Summarise heat exchangers.		
M3.01	Understand the use of heat exchangers in industry.	1	Understanding
M3.02	Illustrate major types of heat exchangers in industry.	6	Understanding
M3.03	Identify the flow patterns and allocation of fluids in heat exchangers.	3	Understanding
M3.04	Compute heat transfer area for heat exchangers.	6	Applying
Contents: Heat exchangers – Uses of heat exchangers in industry Illustrate double pipe heat exchanger - Shell and tube heat exchanger - U tube heat exchanger - Floating head heat exchanger - Plate type heat exchanger and finned tube heat exchanger– Identify the functions of tube sheet, tie rod and baffles in heat exchangers – Identify the types of baffles. Flow patterns in heat exchangers - Parallel flow – Counter current flow – Cross flow heat exchangers – Single pass and multi pass heat exchangers – Naming of heat exchanger based on number of passes– Illustrate flow pattern in multi pass heat exchangers- Outline the routing of shell and tube side fluids in heat exchangers. Temperature distribution in heat exchangers, LMTD - Estimation of heat transfer area and length of tubes for heat exchangers (LMTD method) – Effect of fouling in heat exchangers.			

CO4	Summarise evaporators.		
M 4.01	Understand the use of evaporators in industry.	1	Understanding
M 4.02	Illustrate different types of evaporators.	6	Understanding
M 4.03	Compute heat transfer area for an evaporator.	4	Applying
M 4.04	Illustrate multiple effect evaporators.	4	Understanding
	Series Test – II		

Contents:

Evaporators – Uses of evaporators in industry.

Constructional Details and Working Of Evaporator - Types of evaporators– Basis of classification – Horizontal tube – Vertical tube – Climbing film – Falling film – Forced circulation evaporators – Examples of application of each evaporator in industries.

Material and heat balance equation for single effect evaporators – Simple problems – Calculation of heating area – Define: capacity and steam economy.

Multiple effect evaporation – Principle of multiple effect evaporation – Methods of feeding – Advantages of multiple effect evaporation - Temperature distribution in multiple effect and effect of boiling point elevation.

Text / Reference

T/R	Book Title/Author
T1	Heat Transfer; K A Ghavane
R1	Heat transfer and its application; Binay K Dutta
R2	Introduction to Chemical Engineering; Water.L.Badger
R3	Unit operation of Chemical Engineering; Warren.L.McCabe andSmith
R4	Heat transfer; D.D.Tiwari

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/103/103/103103032/